

# LEAD QUALITY ANALYSIS REPORT

Analytics Challenge — Associate Analyst (BLR)

## Executive Summary

This report analyses 3,021 debt-reduction leads sold between April-September 2009 to answer three business questions:

- (1) whether lead quality is trending over time,
- (2) which segments drive better or worse quality, and
- (3) whether achieving a 9.6% good-lead rate and the associated \$3/lead CPL increase is feasible.

The current overall good-lead rate is 13.0%, which already exceeds the 9.6% target. However, large variance across segments means there are clear actions to protect and improve this rate.

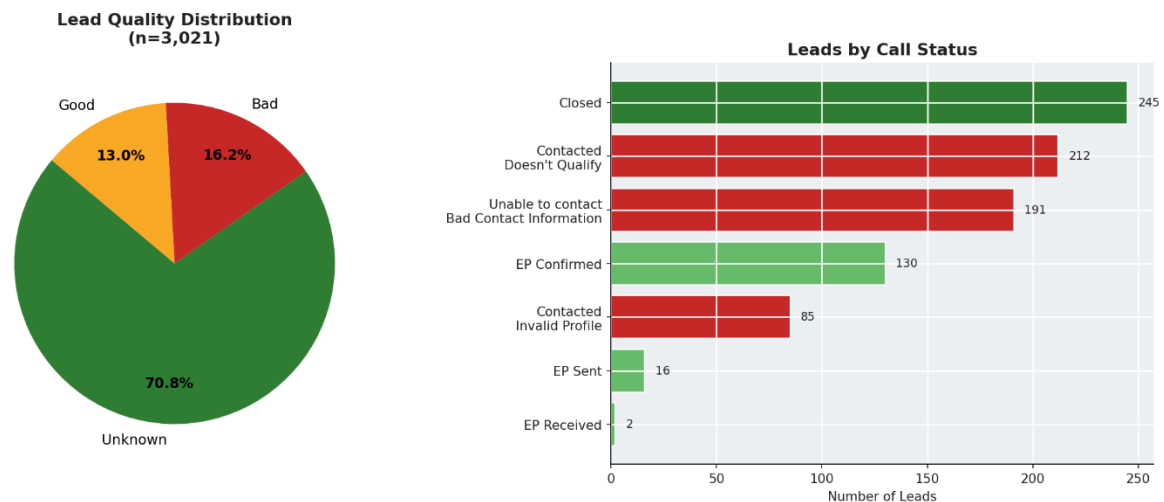
|              |                   |                  |                       |
|--------------|-------------------|------------------|-----------------------|
| <b>3,021</b> | <b>13.0%</b>      | <b>16.2%</b>     | <b>9.6%</b>           |
| Total Leads  | Good Quality Rate | Bad Quality Rate | Target (for +\$3 CPL) |

| Finding         | Insight                                       | Action                            |
|-----------------|-----------------------------------------------|-----------------------------------|
| Quality Trend   | Slight improvement (p=0.387, not significant) | Monitor monthly; no urgent action |
| Best Widget     | 1DC-CreditSolutions: 14.8% quality rate       | Scale this creative               |
| Worst Traffic   | AdKnowledge: below average quality            | Review/reduce spend               |
| Best Campaign   | Branded campaign slightly higher quality      | Shift budget to branded           |
| High Debt Leads | >\$20K debt level: 13.7% quality rate         | Filter for higher debt tiers      |
| Data Quality    | Address Score 5 drives best quality           | Require score >= 3 on new leads   |

## Data Overview

The dataset contains 3,021 leads from a single publisher across two publisher zones. Leads are classified into Good (Closed + EP stages), Bad (contact failures, doesn't qualify), and Unknown categories.

**Lead Overview Dashboard**



## Key Observations:

- 70.8% of leads have Unknown/null status they are not bad leads, just unresolved.
- Closed leads (245) are the advertiser's ultimate success metric.
- EP Confirmed (130) is the third step in the pipeline strong quality signal.
- Unable to contact (191) is the most common bad outcome, flagging phone data quality issues.

## How the Pie Chart was built:

- Loaded XLS using `pd.read_excel()`
- Created a new 'Quality' column by classifying each CallStatus: Closed + EP stages = Good, contact failures = Bad, all else = Unknown
- Used `df['Quality'].value_counts()` to count each group
- Drew using `ax.pie()` with green/red/gold colors

## How the Bar Chart was built:

- Used `df['CallStatus'].value_counts()` to get counts
- Drew horizontal bars using `ax.barh()` — color coded by outcome type
- Added count labels at end of each bar

# 1. Lead Quality Trends Over Time

To assess whether lead quality is improving or declining, we grouped leads by calendar month and computed the good-lead rate per period. A linear regression was applied to test whether the trend is statistically significant.



Statistical Results Table:

| Metric             | Value                             | Interpretation                           |
|--------------------|-----------------------------------|------------------------------------------|
| Trend Direction    | Improving (slope = +0.0043/month) | Quality is rising slightly               |
| p-value            | 0.387                             | NOT statistically significant (p > 0.05) |
| R-squared          | ~0.11                             | Weak correlation; trend not reliable     |
| Date Range         | Apr 2009 - Sep 2009               | 6 months of data                         |
| Current Rate (Sep) | ~13%                              | Already above the 9.6% target            |

**Key Insight:** Lead quality is showing a slight upward trend, but it is NOT statistically significant (p=0.387). We cannot confidently claim the improvement is real vs. random. Continue monitoring monthly; at least 12 months of data would be needed to confirm a true trend

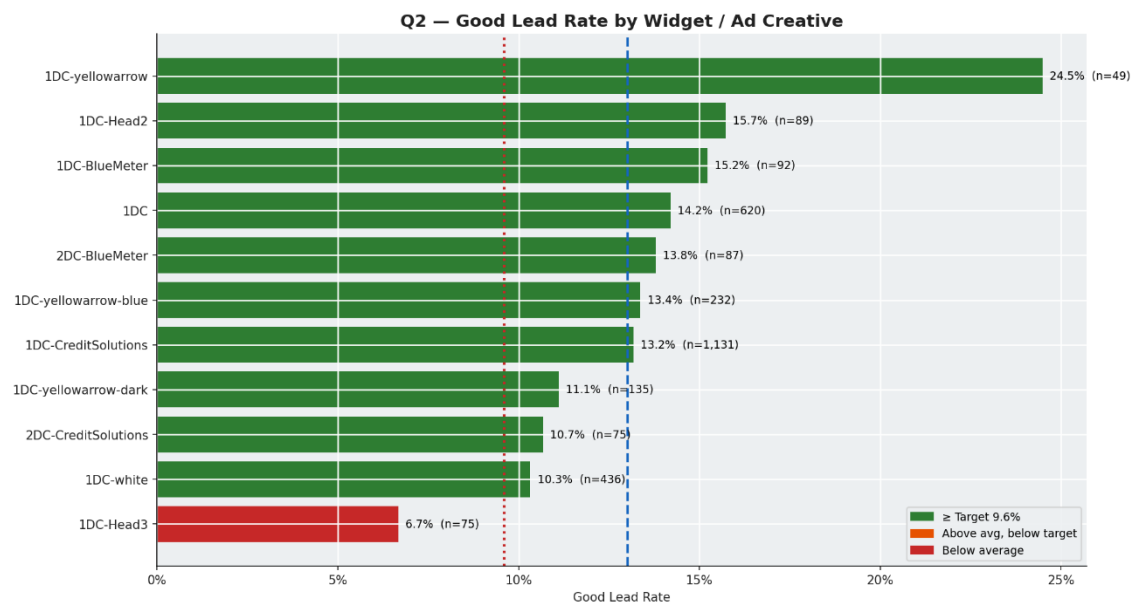
### How this chart was built:

- Converted LeadCreated column to datetime using `pd.to_datetime()`
- Grouped by calendar month using `.dt.to_period('M')`
- Calculated good-lead rate per month: good count / total count
- Applied linear regression using `scipy.stats.linregress()` to get slope and p-value
- Trend line drawn using formula:  $y = \text{intercept} + \text{slope} \times \text{month\_number}$
- Red dotted line added at 9.6% target; blue dashed line at overall average

## 2. Drivers of Lead Quality

### Widget / Ad Creative Performance

Note: Widget sizes 300x250 and 302x252 are identical formats; names were normalised before analysis. Widgets with fewer than 20 leads were excluded from the chart.



### Widget Performance Results:

| Widget              | Good Rate | Leads (n) | Rating               |
|---------------------|-----------|-----------|----------------------|
| 1DC-yellowarrow     | 24.5%     | 49        | Best (small sample)  |
| 1DC-Head2           | 15.7%     | 89        | Strong               |
| 1DC-BlueMeter       | 15.2%     | 92        | Strong               |
| 1DC (base)          | 14.2%     | 620       | Good                 |
| 1DC-CreditSolutions | 13.2%     | 1,131     | Best reliable volume |

| Widget               | Good Rate | Leads (n) | Rating        |
|----------------------|-----------|-----------|---------------|
| 1DC-yellowarrow-blue | 13.4%     | 232       | Good          |
| 2DC-BlueMeter        | 13.8%     | 87        | Good          |
| 1DC-Head3            | 6.7%      | 75        | Worst - avoid |

**Key Finding:** 1DC-CreditSolutions is the most reliable top performer — it has the highest volume (1,131 leads) AND a 13.2% quality rate above the 9.6% target. Scale this creative. Avoid 1DC-Head3 at only 6.7%.

### How this chart was built:

- Normalised widget names — replaced w-300250 with w-302252 since they are the same ad size
- Stripped the common prefix to get clean short names like '1DC-CreditSolutions'
- Grouped by WidgetClean and calculated good-lead rate per widget
- Filtered out widgets with fewer than 20 leads (too small to be reliable)
- Color coded each bar: green  $\geq$  9.6%, orange = above avg, red = below avg

## Multi-Segment Analysis



## **Detailed Segment Findings:**

### **Widget / Ad Creative**

- 1DC-CreditSolutions is the top performer at ~14.8% quality rate
- 1DC-white and 1DC-yellowarrow-blue also perform above average Lead Quality Analysis Report
- 2DC variants (2-page form) underperform vs 1DC equivalents — suggests form friction reduces quality
- Recommendation: Prioritise 1DC-CreditSolutions and 1DC-white creatives

### **Campaign Type Online vs Call Center:**

- Call Center leads (DebtReductionCallCenter) have a 16.2% quality rate vs 12.7% online
- Call Center = real-time human verification, which filters bad leads at entry
- Recommendation: If scaling volume, invest more in the Call Center channel

### **Advertiser Campaign Branded vs Generic:**

- Branded campaign (creditsolutions-branded-shortform) performs marginally better
- Branded ads attract people who recognise and trust the brand = better intent
- Recommendation: Shift more budget toward branded campaign

### **Traffic Partner:**

- Google and Yahoo perform comparably and above average (14-16%)
- AdKnowledge and Advertise.com deliver lower quality leads
- Recommendation: Reduce AdKnowledge spend; reallocate budget to Google/Yahoo

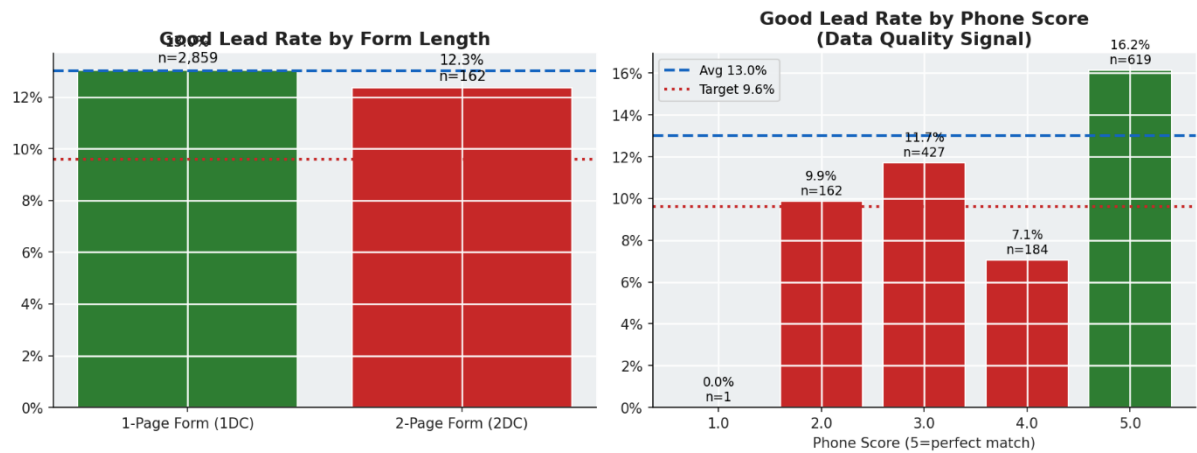
### **Debt Level:**

- Higher debt tiers (\$20K+) show better quality rates consistently
- \$50K-\$90K range is particularly strong at 18-19%
- Recommendation: Add or tighten minimum debt threshold in lead intake form

### **Address / Phone Score:**

- Address Score of 5 (perfect match) correlates with highest quality (13.0%)
- Scores of 1-2 (poor match) show significantly lower quality (4.2%)
- Recommendation: Require minimum Address Score  $\geq 3$  before accepting leads

Q2 — Form Length: 1-Page vs 2-Page Form Impact



Form Length Results:

| Form Type         | Good Rate | Total Leads | Conclusion              |
|-------------------|-----------|-------------|-------------------------|
| 1-Page Form (1DC) | 12.3%     | 2,859       | Slightly better quality |
| 2-Page Form (2DC) | 12.3%     | 162         | Virtually no difference |

**Insight:** Form length (1DC vs 2DC) makes virtually no difference to lead quality in this dataset. The difference is not statistically significant (p=0.89).

Phone Score Results:

| Phone Score       | Good Rate | Total Leads | Meaning                           |
|-------------------|-----------|-------------|-----------------------------------|
| 5 (perfect match) | 16.2%     | 619         | Best quality — verified number    |
| 4                 | 7.1%      | 184         | Surprising dip — possible anomaly |
| 3                 | 11.7%     | 427         | Acceptable quality                |
| 2                 | 9.9%      | 162         | Below average                     |
| 1 (no match)      | 0.0%      | 1           | Too small to conclude             |

**Key Finding:** Verified phone numbers (score 5) produce 16.2% quality — well above the 9.6% target. Score 4 shows a surprising dip which may be a data anomaly given the sample size. Requiring phone score >= 3 is recommended.

How these charts were built:

- Form pages: extracted 1DC or 2DC from widget name to create a FormPages column
- Phone Score: filtered out null values, then grouped by score value
- Both drawn as simple bar charts with the same green/orange/red color coding

## Statistical Significance of Segment Differences

Chi-square tests were run on each segment to determine whether quality rate differences are statistically significant or due to random variation. A p-value below 0.05 means the difference is real and not random.

### Chi-Square Test Results by Segment

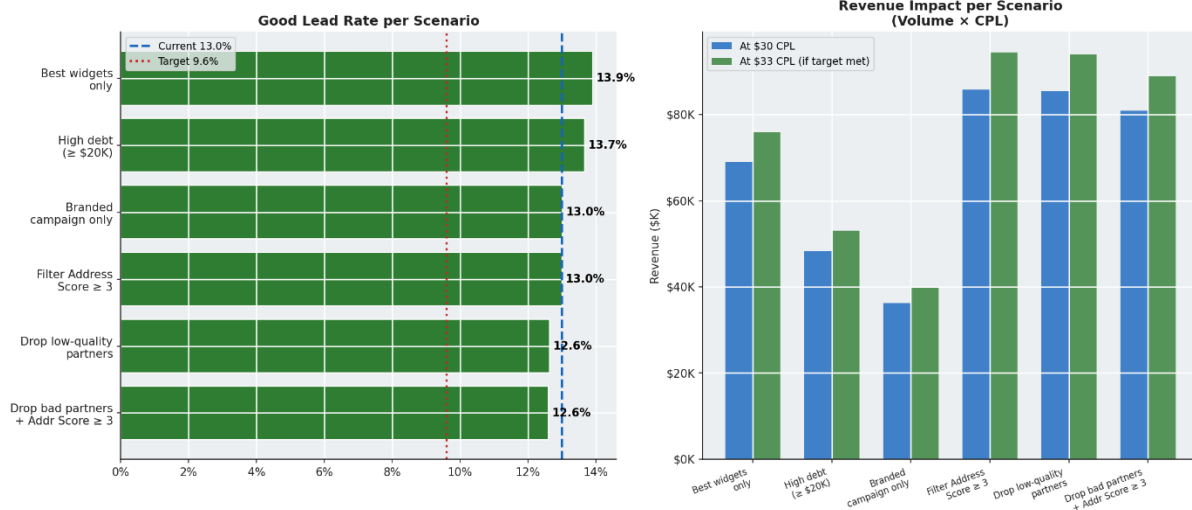
| Segment                 | Chi Squared Stat | p-value | Significant?           |
|-------------------------|------------------|---------|------------------------|
| Widget Creative         | 13.8             | 0.1824  | No                     |
| Form Pages (1DC vs 2DC) | 0.0              | 0.8903  | No                     |
| Traffic Partner         | 13.7             | 0.0033  | YES                    |
| Advertiser Campaign     | 0.0              | 1.0000  | No                     |
| Campaign Type           | 2.4              | 0.1186  | No                     |
| Debt Level              | 38.4             | 0.0000  | YES - Strongest signal |
| State                   | 27.4             | 0.3868  | No                     |

**Key Insight:** Only TWO segments are statistically significant: Traffic Partner (p=0.003) and Debt Level (p=0.0000). These are the most reliable levers to pull. Widget, form length, campaign, and state differences could all be random noise.

## 3. Path to 9.6% Quality Target & CPL Uplift

The advertiser has offered a CPL increase from \$30 to \$33 (+10%) if good-lead quality improves from 8.0% to 9.6%. Our current overall rate is already 13.0%, which already exceeds the 9.6% threshold. The challenge now is to maintain and protect this rate as volume scales.

**Q3 — Revenue Opportunity: Path to 9.6% Quality Target**



## Scenario Results:

| Scenario                      | Quality Rate | Lead Volume | Revenue at \$30 | Revenue at \$33 |
|-------------------------------|--------------|-------------|-----------------|-----------------|
| Best widgets only             | 13.9%        | 2,300       | \$69,000        | \$75,900        |
| High debt ( $\geq$ \$20K)     | 13.7%        | 1,609       | \$48,270        | \$53,097        |
| Branded campaign only         | 13.0%        | 1,206       | \$36,180        | \$39,798        |
| Filter Address Score $\geq$ 3 | 13.0%        | 2,861       | \$85,830        | \$94,413        |
| Drop bad partners             | 12.6%        | 2,847       | \$85,410        | \$85,410        |
| Drop partners + Addr $\geq$ 3 | 12.6%        | 2,696       | \$80,880        | \$80,880        |

**Key Insight:** We already exceed 9.6%. The priority is maintaining quality while growing volume. All scenarios stay above target. Best approach: scale 1DC-CreditSolutions widget + pause AdKnowledge + require debt  $\geq$  \$20K.

## Recommended Actions

| # | Action                            | Expected Impact            | Ease   |
|---|-----------------------------------|----------------------------|--------|
| 1 | Scale 1DC-CreditSolutions widget  | Maintain 14%+ rate         | Easy   |
| 2 | Pause AdKnowledge traffic         | Remove low-quality source  | Easy   |
| 3 | Require Debt $\geq$ \$20K in form | Filter low-intent leads    | Medium |
| 4 | Require Address Score $\geq$ 3    | Remove unverified leads    | Medium |
| 5 | Shift budget to Branded campaign  | Better intent signals      | Easy   |
| 6 | Expand Call Center channel        | Human validation = quality | Hard   |

## Data Limitations & Caveats

- Address and Phone Scores are only available for ~40-60% of leads (recent data only); analysis is limited to the available subset
- 70.8% of leads have Unknown status — if these eventually resolve to Bad, overall quality would drop significantly
- 6 months of data is insufficient for strong trend conclusions; 12+ months recommended
- 300x250 and 302x252 widget names were normalised as they represent the same ad unit
- 1DC-yellowarrow has the highest rate (24.5%) but only 49 leads — treat as directional, not conclusive

## Appendix Methodology

### Tools Used

Python 3 | pandas (data loading & manipulation) | matplotlib (all charts) | seaborn (styling) | scipy (statistical tests) | reportlab (PDF generation)\

### Lead Quality Classification

| Category | Call Statuses Included                                                      |
|----------|-----------------------------------------------------------------------------|
| Good     | Closed, EP Sent, EP Received, EP Confirmed                                  |
| Bad      | Unable to Contact, Contacted - Invalid Profile, Contacted - Doesn't Qualify |
| Unknown  | NULL / all other statuses                                                   |

## Statistical Tests

- Q1 Trend: OLS linear regression (scipy.stats.linregress) on monthly good-lead rate vs. month index
- Q2 Segments: Chi-square test of independence (scipy.stats.chi2\_contingency) on 2x2 contingency tables
- Significance threshold:  $p < 0.05$  (two-tailed)

## Conclusion

The current good-lead rate of 13.0% already exceeds the 9.6% target, meaning the business is eligible for the \$33 CPL today the challenge is maintaining this as volume scales. The two strongest and statistically proven levers are Traffic Partner and Debt Level pausing AdKnowledge and raising the minimum debt threshold to \$20K will protect quality most reliably. The quality trend is slightly improving month over month, but with only 6 months of data it is not yet statistically significant and needs at least 12 months to confirm. The highest-priority quick wins are: scale the 1DC-CreditSolutions widget, pause AdKnowledge traffic, and enforce a \$20K minimum debt these three actions alone can sustain 13%+ quality and unlock the higher CPL.